
Flexible Re-Cycling and High-Technology Entrepreneurship

Homa Bahrami
Stuart Evans

High-technology firms in Silicon Valley prosper in a constantly changing environment of incessant novelty and innovation, which fuels their volatile, occasionally spectacular, growth trajectories. This unique domain is characterized by fleeting opportunities, shifting customer preferences, cascades of technological innovations, brutally short product life cycles, and furious global competition.¹ These forces collectively precipitate never-ending streams of "kaleidoscopic" change that can stimulate the rapid growth of an enterprise and also instigate their sudden demise.

A firm's chances of success may be small in this highly combative arena, yet this unique ecosystem has spawned several of the fastest growing firms in the recent history of U.S. business while maintaining a maverick, entrepreneurial spirit.² Silicon Valley is unique in the scale and success of creating a high-technology cluster in an environment, which, only three decades ago, was resplendent with orchards. The entrepreneurial spirit and commitment to innovation have so far proven effective in producing pioneering products and high-value-added jobs³. Furthermore, Silicon Valley firms seem to be pioneering a flexible form of enterprise conceived for the fast-moving, global information age.⁴

Their managerial recipes deviate from conventional business practices, and their business and organizational systems implicitly question several core assumptions that are embedded within the traditional body of management theory and organizational practice.⁵ One central assumption, embedded within the conventional model of the business enterprise, is that of "permanence." This deeply rooted, ontological assumption pervades economic theory in the search for equilibrium. It resonates in the quest for sustainable competitive advantage or enduring strategies and is viewed as a critical ingredient of organizational

effectiveness.⁶ Many studies use the notion of organizational death as an indicator of “fit or misfit” within a particular domain, with important consequences for the analysis of business firms, industrial arenas, and national economies.⁷

The broadly held view that the ultimate goal of any enterprise—and, by implication, its key measure of success—is to exist in perpetuity is partly at odds with Silicon Valley’s experience.⁸ Paradoxically, while high-technology firms experience a high failure rate,⁹ Silicon Valley continues to thrive and prosper.¹⁰ In this domain, the demise of one firm invariably leads to the formation of others, directly and indirectly.¹¹ This process of “flexible re-cycling” can result in novel re-configurations of knowledge and human capabilities, allowing new firms to rise from the ashes of failed enterprises.

Questioning the universal validity of the “permanence” assumption for all high-technology entities does not preclude many firms from enduring over time.¹² It does, however, raise the specter of alternative models of success and business effectiveness—especially for clusters of knowledge-based high technology and other highly dynamic sectors.¹³ Ephemeral firms are an integral part of renewal in Silicon Valley, contributing through a flexible re-cycling process to the continued success of the ecosystem, so that indeed there is life after death in this knowledge-based arena.

Silicon Valley is more than a random cluster of technology firms located in a well-defined geographic context. In much the same vein as a natural ecosystem, Silicon Valley’s growth and success can be attributed to the incessant formation of a multitude of specialized, diverse entities which feed off, support, and interact with one another. Constituents of this ecosystem include venture capitalists, a global talent pool of knowledge professionals, universities and research institutes, a sophisticated service infrastructure, as well as many customers, lead-users, and early adopters of new technologies.

While several studies have examined specific industries¹⁴ or sectors, few have considered this unique “ecosystem.”¹⁵ This article draws on several field studies of the high-technology arena which were conducted in Silicon Valley from 1982 to 1993.¹⁶ It explores the Silicon Valley ecosystem and describes some of its major components. While many firms in the dynamic high-technology arena may be ephemeral, the Silicon Valley ecosystem is renewed and nurtured by a continuous process that we term “flexible re-cycling.” This ecosystem provides an anchor of stability within which incumbent firms and new start-ups can flourish and become a source of innovation and employment, and yet remain sufficiently flexible to accommodate the constant stream of kaleidoscopic changes.

Historical Antecedents Of Silicon Valley

Clusters of firms in related industries have historically coalesced around a critical mass of business activity. During the 19th century, many firms in

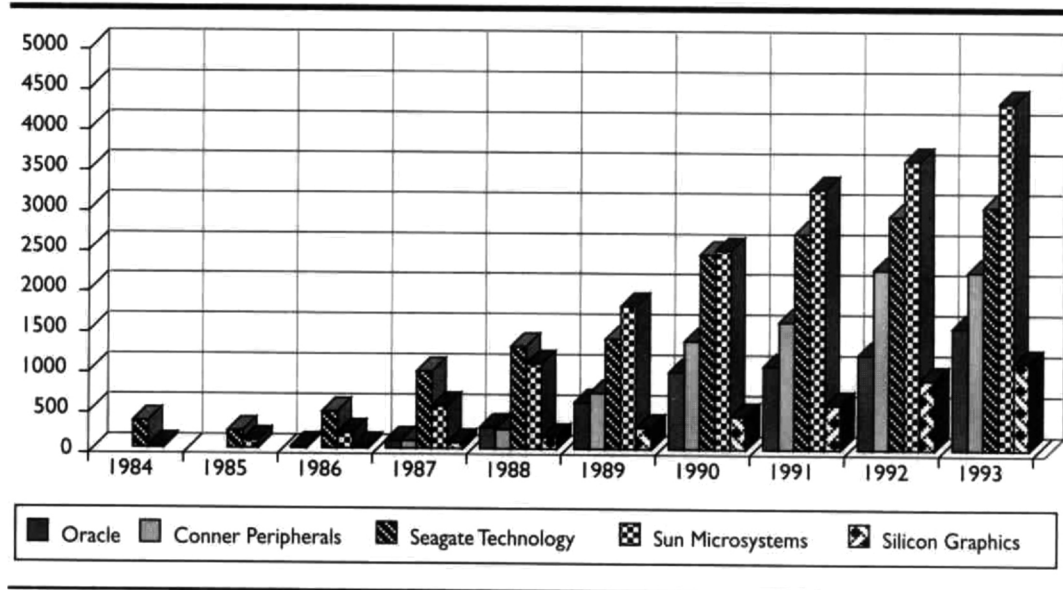
Birmingham, UK, clustered around the critical mass of expertise in what is known in the vernacular as "metal bashing." During this century, the automobile sector amassed around Birmingham (U.K.), Detroit (U.S.), and Stuttgart (Germany). In the City of London, the financial industries have evolved around the famous "square mile." Similarly, Italy's textile industry has coalesced around the city of Prato.¹⁷ High-technology industries of the information era also appear to conform to this tendency.

Several regional technology clusters have sprouted around the U.S. in recent years—including Austin's "Silicon Hills," Seattle's "Technology Corridor," Illinois's "Silicon Prairie," New Jersey's "Princeton Corridor," San Diego's "Golden Triangle," and Utah's "Software Valley."¹⁸ Scotland's "Silicon Glen" has attracted several technology-based companies, and Singapore has become a center for disk drive and computer sub-systems. Additionally, a number of government-sponsored science park initiatives, such as France's Sofia Antipolis, Taiwan's Hsinchu,¹⁹ and U.K.'s Cambridge,²⁰ have also attempted to induce a critical mass of high-technology firms. The best-known and the largest cluster of high-technology firms in the electronics industry is located in California's Silicon Valley.

Silicon Valley sprouted spontaneously when several complementary forces gelled in the late 1960s, producing a mutually supportive spiral of entrepreneurship and innovation, which resulted in the formation of a critical mass of high-technology firms. There was no singular event or grand plan that in-and-of-itself led to the meteoric rise of "Silicon Valley."²¹ Instead, a series of independent events, coupled with fortuitous timing, transformed a regional agricultural community into a global complex of technological innovation and high-technology entrepreneurship. This versatile ecosystem enables fledgling firms to draw on specialized support services and the diverse pool of talent so they can concentrate on their unique expertise. Successful firms in Silicon Valley have grown at a phenomenal pace. Figure 1 illustrates the growth trajectories of selected companies.²²

From an technological perspective, the parallel development of two major innovations forged the critical building blocks that underpinned the rapid growth of Silicon Valley in the 1960s and 1970s. The first, and the most well-known of these, was the commercial development of the transistor at AT&T's Bell Labs in 1948. The second, often overlooked, was the development of disk drives or information storage technology using magnetic recording techniques. Using tape in 1953 and magnetic disks by 1957, the technology was developed by IBM scientists at its Los Gatos Laboratory.²³

The transistor was commercialized by Shockley, one of the Bell Lab scientists, who set up Shockley Semiconductor Laboratories in Palo Alto during the mid-1950s. His working team of eight scientists became the founding nucleus for the growing West Coast semiconductor industry. This team left Shockley Lab in 1957 and founded Fairchild Semiconductor. Further advances in semiconductor technology and the emergence of a major market in the defense industry helped

FIGURE I. Selected Company Growth Trajectories 1984-1993 Annual Revenues (\$ Million)

launch many spin-offs, largely out of Fairchild, during the 1960's—among them National Semiconductor, AMI, Advanced Micro Devices, and Intel.

During the same period, IBM set up its research unit in Los Gatos with the aim of producing technical breakthroughs and innovative products. The development of Winchester disk drive²⁴ later led to the formation of a multi-billion dollar industry in Silicon Valley.²⁵ Several members of the IBM team were later responsible for the founding of Memorex and other well-known disk drive and peripherals firms such as Shugart, Seagate, Conner Peripherals, Adaptec, Auspex, Maxtor, and Quantum (as shown in Figure 4). Much overlooked by commentators, the disk drive industry has been a constant source of innovation and international dominance for U.S. companies.²⁶

Stanford University's Dean of Engineering, Frederick Terman, played a crucial role during the early years by forging a close working relationship between the engineering school and technical firms in the area. The formation of the Stanford Industrial Park in 1951 was an additional boost by providing a mechanism for transferring technology from University laboratories to the nearby commercial firms.²⁷ During the 1960s, the Park became an attractive location for growing electronics companies—expanding steadily from 32 firms in 1960, to almost 70 by 1970.²⁸

Boosted by California's unique pioneering spirit, these critical building blocks formed the foundation of Silicon Valley's entrepreneurial setting. Since the days of the Gold Rush, California had attracted the innovative and the ambitious.²⁹ It is hardly surprising that many entrepreneurs who felt the need to challenge the *status quo* and to break with tradition found a conducive home in California. By the mid-1960s, a critical mass of technology companies had

been established in the South San Francisco Bay area, and Santa Clara Valley had been transformed from prune yards and orchards into a large Petri dish whose nutrient base was an active experimentation and information exchange system for technological know-how.

Constituents of the Silicon Valley Ecosystem

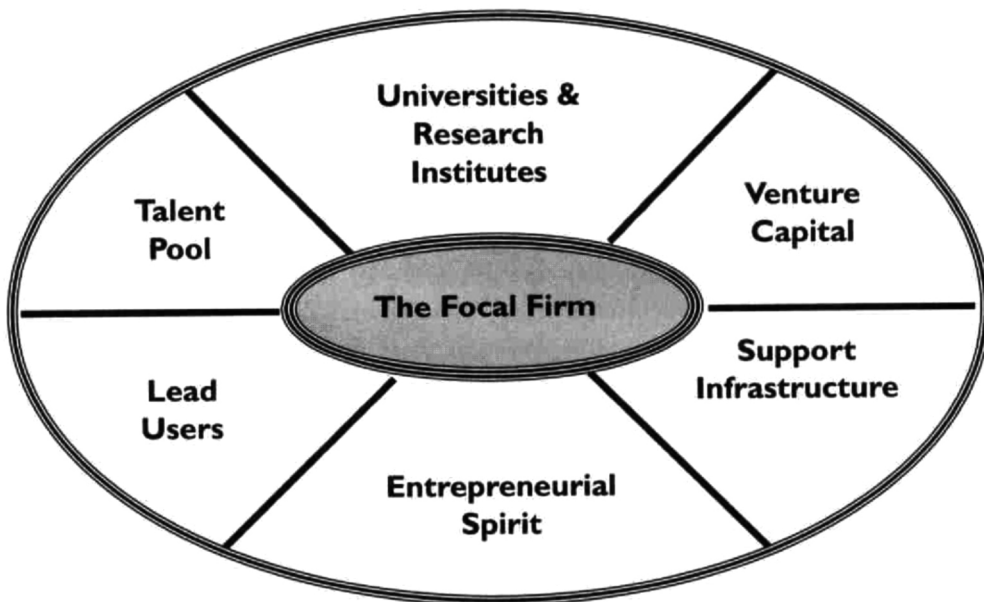
There are several components—each playing different and mutually supportive roles—of Silicon Valley's versatile ecosystem (see Figure 2).

Universities and Research Institutes

The universities and corporate and government research institutes are the most readily identifiable and perhaps the most publicized component. With a strong technological orientation, these institutions train young engineers who make up the pool of entrepreneurs and professionals in Silicon Valley. Acting as a source of pre-commercialization stage technologies, it may be argued that these institutions are the nutrient base of the ecosystem.

Universities are also a catalyst for informal networking amongst future entrepreneurs. The bonds which are forged between the students play a crucial role in the process of firm formation. For example, Sun Microsystems was

FIGURE 2. Constituents of the Ecosystem



formed by two Stanford MBA students and a graduate student from the Stanford Engineering School who combined forces with a doctoral candidate from Berkeley's Computer Science Department. The name of the company is an acronym of the Stanford University Network around which the company coalesced. All four founders of ROLM, were graduate students at the Stanford Engineering School and had previously attended Rice University.

Once in a while, faculty members also start companies. For example a professor at Stanford Medical School co-founded Nellcor, a medical electronics company, which developed an indispensable aid for anesthesiologists. Silicon Graphics was founded by Jim Clark from Stanford's Computer Science Department. Cromemco, the pioneering computer company in the 1970s, was founded by two professors from the Engineering School. The nascent biotechnology industry is populated by a number of firms founded by eminent scientists. Many faculty members are also advisors, consultants, and board members of the growing companies.

Corporate research institutes have played a crucial role in this process of generating critical technological know-how and scientific progress. Xerox's Palo Alto Research Center or PARC, was responsible for the development of local area networks. Bob Metcalfe—the co-inventor of Ethernet at PARC—left Xerox to form 3Com in order to commercialize the technology. Chuck Greschke and John Warnock invented Postscript at Xerox PARC and then spun-off to form Adobe.

Many innovations have emanated from these research institutes without their direct involvement in commercializing the technology. For example, the optical disk drive, the mouse pointing device, and the magnetic ink character system for bank checks were all initially developed at the Stanford Research Institute. The interchange of faculty between the universities and research institutes has also played an important role in the process of innovation and information exchange.³⁰

Venture Capital Community

The venture capital community is a crucial component that provides financial resources to start and grow new entities. Although it is easy to overstate the role of venture capital in creating new companies, the community is responsible for more than simply investing in fledgling companies. Venture capitalists provide valuable management know-how, particularly in the embryonic stages of a firm's development. Furthermore, armed with a network of contacts, they often recommend individuals to augment the management team of a start-up.

Many venture capitalists are, in the main, responsible for funding companies beyond the initial "seed" stage. Private individuals typically provide the seed funding for entrepreneurs to get started, before venture capital is obtained.³¹ Typically, after seed funding, a firm will require four or five rounds of financing

to build manufacturing capacity, forge distribution channels, develop follow-on products, and engage in global expansion. The principal role of the venture capitalists is to provide the funds necessary to grow the enterprise into a sizable business and to put it in a position to undertake an initial public offering or be acquired by another company.³²

Start-ups require several rounds of financing before generating sufficient revenue and earnings growth to undertake an initial public offering (IPO). The initial group of venture capitalists, particularly the lead investor in the consortium, typically assist with follow-on financing and the IPO process.³³ Moreover, the relationships that are forged between entrepreneurs and venture capitalists typically endure over and above any one start-up. In sum, the roles played by venture capitalists are:

- several rounds of funding;
- management know-how;
- network of contacts to augment the team;
- follow-on financing;
- in certain cases, funding the entrepreneurs' next start-up.

A Sophisticated Service Infrastructure

A sophisticated service infrastructure allows start up firms to focus on their chosen steeple of expertise, rather than dissipate their energies across a broad range of peripheral or supporting activities. Contract manufacturing services are available to develop prototypes, or engage in high volume or "peak load" manufacturing of sub-systems and finished goods. Specialized public relations firms provide assistance with strategic marketing, product packaging, trade shows, company logo's and other collateral. Accounting firms have specialized high-technology practices and services designed for start-up and high-technology companies.

Executive search firms are used extensively by start-up and maturing firms in order to augment management teams, and to recruit new talent. Real estate firms have expertise in the provision of facilities, especially designed for high-technology firms. For example, some firms may require clean rooms or highly purified water supplies; others set out to create a "campus"-like environment for their professional employees.³⁴

Law firms also play a crucial role in the creation of new ventures. A handful of prominent law partnerships have grown in Silicon Valley by specializing in high-technology services. In addition to litigation, they perform several major functions, including:

- initial incorporation, company name search,
- stock allocations,
- patent filings,
- alliance agreements,

- public offering prospectus,
- SEC filings,
- acquisition agreements.

At the outset, a new start-up is typically offered favorable fee structures, in the anticipation that substantial legal assistance will be required as it undertakes additional rounds of financing and eventually, four or five years downstream, may be acquired or embark on an initial public stock offering. Senior partners typically forge long-standing relationships with the venture capital community and often refer entrepreneurs to venture capitalists who have expertise or prior experience with a specific type of venture or industry. This further underscores the role of personal relationships in the dynamics of the ecosystem.

The function of this supporting infrastructure is the timely provision of a wide range of specialized expertise so that things can get done quickly and effectively. If a firm needs to prototype an integrated circuit to test a new design, it can be fabricated in a matter of days; if it needs a booth for a trade show, it can be put together over a weekend; if it requires specialized advice, it can be provided by a phone call.

Talent Pool

A diverse talent pool of professionals has gravitated toward Silicon Valley from all over the globe.³⁵ Many come to attend the universities and remain in the area; others are recruited to work for Valley-based companies overseas and transfer back to the home base. Since many high-technology firms are global from their inception, this diversity is particularly helpful in developing global operations quickly and efficiently.

Entrepreneurial Spirit

Finally, an intangible, but nonetheless critical ingredient of the Silicon Valley ecosystem is the pioneering spirit and relentless work ethic. The entrepreneurial culture was initially born out of a Californian history of pioneers making the perilous journey over the Rocky Mountains, coupled with the legacy of the Gold Rush. The culture—ingrained over time—is characterized by hard work, inspiration, and by doing one thing very well.

Entrepreneurs exhibit many of the qualities of the early pioneers. They are prepared to take enormous risks, innovate in areas that most say can't be done, work incredibly long hours over extended periods of time, and even suffer personal problems, all for the quest of developing a product or building an enterprise. They have passion and bring a singular focus to their projects. As Schumpeter said, "The inventor produces ideas, the entrepreneur gets things done."³⁶

The Ecosystem: Flexible Recycling

In order to understand the modus operandi of Silicon Valley it is instrumental to examine it both at the level of ecosystem as a whole and at the level of the individual firm. Once the broader context is set, it is possible to better grasp how the individual firms function in a symbiotic relationship with this unique ecosystem.

Incessant Creation of New Ventures, Rapid Recycling of Failures

Each year, approximately 300 new firms are funded by venture capital in Silicon Valley.³⁷ Notably, more firms are self-funded by founding teams or with investment assistance from private individuals colloquially known as "angels." As indicated in Table 1, new firms are founded in several different ways.

The notion of spin-offs is well documented in the management literature and plays a significant role in the creation of new high-technology firms. Indeed,

spin-offs, in the traditional sense of the term, do represent a significant source of new firm formation in Silicon Valley. Nevertheless, the commonly accepted definition of a spin-off—defined as individuals from one company forming a new entity—is not representative of the majority of Silicon Valley firms. Instead, founding teams typically get together from different firms and backgrounds to form new enterprises. The founding teams of these

TABLE 1. Categories of Entrepreneurial Ventures

Categories	Examples
Spin-offs	Adobe Systems, SynOptics, 3Com, General Magic, Ultra
A Group of Friends/Colleagues	H-P, ROLM, Sun
Scientists	Genetech, Synaptics, Nellcor, SGI
Hobbyist/Enthusiast	Apple, MacroMedia
Corporate/VC-induced	EO-GO, Starent, Trilogy
2nd Time Entrepreneurs	Seagate, Conner, 3DO, Auspex, Echelon, NeXT
Re-starts	SuperMac, Centigram

hybrid spin-offs may initially get together at networking forums, such as special interest group meetings, and universities, or through previous relationships as customers, vendors, or working associates.

New start-ups are also founded by technology hobbyists and enthusiasts who may develop a product, initially for their own use. The most famous of garage start-ups, Apple Computer, came into being in this manner,³⁸ as did MacroMind (now MacroMedia) the multimedia software firm. There are a handful of "repeat" entrepreneurs. In the disk drive industry, Alan Shugart parted company with Shugart Associates and, although excluded from competing in the disk drive industry for a number of years, co-founded Seagate Technology with another Shugart co-founder, Finis Conner, in 1979. Following a downturn in the disk drive industry in 1985, Finis Conner left Seagate to form

his own company, Conner Peripherals, which became one of the fastest growing companies in the U.S. business history.³⁹ Similarly, Larry Boucher, left his first entrepreneurial venture, Adaptec, to found Auspex Systems in 1988.⁴⁰

Yet another type of new high-tech firm can be best described as a "re-start." These are typically ventures which are either acquired by larger firms (and may later flounder) or are venture-backed start-ups whose business models need re-calibration under new management teams. Some may be able to rise out of the ashes of the failing parent or be bought out to achieve independent viability. SuperMac Technology is a good example of this situation. Scientific Microsystems, the now-defunct Input/Output controller firm, acquired SuperMac in an attempt to diversify. When the parent began to flounder, the founding entrepreneur reversed the process. Some years later, SuperMac orchestrated its own public offering, prior to merging with Radius in 1994.

A further source of new enterprises, one which is often discussed in the literature, is that of researchers/scientists wishing to commercialize their ideas. The biotechnology industry is populated by many entrepreneurs who fall into this category, with Boyer and Cohen, the co-founders of Genentech, as the most famous examples. In the electronics industry, Carver Mead, a professor from Cal Tech and Frederico Fagin, founder of Zilog, co-founded Synaptics to develop neural processors. Syntex commercialized the Nobel Prize-winning discovery of the contraceptive pill. Regardless of their origin, the most striking feature of new enterprises in Silicon Valley is the preponderance of entrepreneurial teams with complementary perspectives and diverse capabilities who band together to found high-tech start-ups.

Continuous Inter-Firm Mobility

Significant fluidity exists between the various components of the Silicon Valley ecosystem. Executives move from high-tech companies into venture capital, investment banking, consulting—among others—and the movement may be in the other direction as well. Engineers change jobs with much frequency and may move from one firm to another just "down the road"—without having to move homes or their children's schools. Consultants move on to become managers and executives. Many executives leave the relative security of an established firm to join a smaller firm or even to start their own companies.

There are many examples of this mobility in Silicon Valley. For example, Gene Kleiner, founding partner of Kleiner Perkins, Caufield & Byers, one of the Valley's leading venture capital firms with \$673 million under management in 1992,⁴¹ was previously at Fairchild Semiconductor. Bob Maxfield and Regis McKenna also joined the venture firm after founding ROLM and Regis McKenna, Inc. (the marketing consulting firm) respectively. Dave Crockett, the founder of Pyramid Technology, also moved into the venture capital arena. Conversely, George Sollman moved from Shugart, the disk drive company, to the venture capital community and later became the CEO of a voice-messaging firm, Centigram.

Executives also frequently move from large firms to startups; Cyril Yansouni moved from H-P to Convergent Technology (and to Unisys when the latter was acquired in 1986) and later to Read-Rite (a manufacturer of thin-film heads for disk drives) as its chief executive. Similarly, his former boss, Paul Ely moved from H-P to Convergent, and subsequently into the venture community. Other examples include Ed McCracken from H-P to Silicon Graphics, William Krause from H-P to 3Com, Trip Hawkins from Apple to Electronic Arts and subsequently to found 3DO, Wilf Corrigan from Fairchild to LSI Logic, Noyce and Moore from Fairchild to found Intel.

This high degree of mobility across various parts of the ecosystem is not only restricted to the business community. For example, Stig Hagstrom, who was responsible for one of the labs at Xerox PARC, moved to Stanford University's Material Science Department. Moreover, this mobility is just as frequent among firms which may have been competitors—within prescribed legal constraints. For example, after stepping down as the Chief Operating Officer of Seagate Technology, Tom Mitchell, moved to a similar position at Seagate's rival, Conner Peripherals.

At its core, the Silicon Valley ecosystem functions through a interconnected network of personal relationships. Universities as well as the more established firms such as H-P, may be the initial context which bring people together and facilitate the formation and nurturing of personal relationships. Since relationships typically endure longer than an individual's tenure at any given company, they provide the fulcrum around which the ecosystem can flexibly operate. Both ROLM and Shugart (and many of their spin-offs) were forged by individuals who had known one another for some time. In the semiconductor industry, Shockley Laboratories was the initial setting which brought together the key group of people who later founded Fairchild and then spun off to form Intel, National Semiconductor, and LSI-Logic, among others. Informal networks thus form the hub of activity in Silicon Valley. These non-institutionalized relationships are difficult to conceptualize, yet are crucial in providing anchors of stability in the dynamic context of the ecosystem.

Rapid Transmission and Widespread Diffusion of Information: Emphasis on "Doing" rather than Just "Knowing"

It is exceedingly difficult to keep secrets in Silicon Valley concerning future products and competitive intentions. This is due, in part, to the close proximity of companies, the fast-moving nature of high-technology industries, the high mobility rate of engineers and other professionals, and the frequent formation of partnerships and alliances that cross-pollinate knowledge and ideas. These, together with short product cycles and market windows, imply that the competitive challenge is not just knowing what to do (in terms of next generation products), but to build new products which can find a receptive market.

In high-technology industries, information has a limited time value, regardless of the level of secrecy a firm may be able to keep concerning its future intentions. New products are *alpha*- and *beta*- tested by potential customers, who may also conduct business with a firm's competitors. In addition, pioneering firms typically have to release technical information in order to ensure compatibility of the various technical sub-systems so that related products can interface effectively. Moreover, due to the rapid pace of change, information about products, markets, and competitors quickly becomes obsolete.

Formal and informal exchanges among "techno-evangelists," on electronic bulletin boards and at user group meetings, is another source of information diffusion. These individuals have a passion for their technical interests and often interact at user group meetings or trade shows with others of similar disposition. The ecosystem provides a broad framework within which people can interact and exchange information, ideas, and know-how.

Learning by Doing and by Failing

In Silicon Valley, there is no stigma attached to honest failure, although there is to resting on laurels and not playing the game. Entrepreneurs are measured by what they are currently doing, not by what they did in the past, whether their previous venture was a success or a failure. Since the stigma associated with failure is depersonalized, both start-ups and incumbents can engage in novel experiments in the belief that it is immeasurably better to try something risky and to fail, rather than to wonder about what might have been. Even if a venture fails, entrepreneurs can learn from the experiment and move on to start a successful entity.⁴² The inverse of this proposition also applies in that a previously successful entrepreneur may not be able to repeat that success in a new incarnation.

Attitudes towards failure are critical to the ultimate success of pioneering high-technology companies.⁴³ As several studies have shown, new products in technology-based industries are successively refined as the result of a learning process, in which innovators not only learn by "doing" but also by "failing."⁴⁴ Such "failures" in turn, may result in a re-assessments of the original concepts and the development of new alternatives. The case of Apple III and Lisa—two unsuccessful computer systems introduced by Apple during the early 1980s that gave birth to the highly successful Macintosh—illustrate the point.

This empiricist approach of trial and error captures the essence of "learning-by-doing" in high-technology companies.⁴⁵ This emphasis on continuous recalibration is especially critical when there are no historical precedents or success recipes for a given product or market arena. Even with the most elaborate planning and focus group meetings, it is not clear, ahead of time, how customers are likely to react to a new class of product.

***Inter-Linked Specialization and Complementary Alliances:
Focus on Creating "Steeple" of Expertise***

The Silicon Valley ecosystem achieves flexibility through "diverse specialization," each firm focusing on what it does best and leverage the capabilities of other entities for complementary activities.⁴⁶ For example, a start-up can focus on its technical design capability, while utilizing other entities for prototype development, manufacturing, market research, public relations, advertising, and staffing. Such an infrastructure is robust. Each component is semi-autonomous, enabling the ecosystem as a whole to withstand shocks and perturbations. It is also versatile in that new competencies can be quickly added, enhancing and upgrading its capabilities.

In this context, the ecosystem may be viewed as a constellation of specialized enterprises and complementary alliances. Alliances are forged between individuals when they coalesce into an entrepreneurial team to form a start-up; venture capitalists forge alliances in the form of a syndicate to co-invest in a firm. In addition, a recent trend, which takes the notion of inter-linked specialization and complementary alliances one step further, is for large firms to ally themselves with start-ups at the early stages not just as investors, but also as potential manufacturing, development, or distribution partners.

EO, the now-defunct personal communications start-up, had a number of global partners (including AT&T, Matsushita, Marubeni, among others) only 18 months into the venture's life-cycle, before being acquired by AT&T. 3DO, a multimedia firm founded by Trip Hawkins of Electronic Arts, was initially a joint venture between Time Warner, Matsushita, and Electronic Arts, before embarking on a public offering in May 1993. Similarly, General Magic, the personal communications software company and an Apple spin-off, was initially forged through an alliance between Apple, AT&T, Philips, and Sony. This partnering trend may be a harbinger of the next evolutionary phase of Silicon Valley.⁴⁷

Several high-technology firms have long participated in global alliances. For example, Quantum Corporation, the disk drive firm, has, for some time allied itself with Matsushita as its manufacturing partner so that it can focus on new product development and distribution. Auspex Systems, the innovative network file server company, forged an alliance with Fuji-Xerox in its formative years, involving both investments and distribution agreements. These arrangements are particularly helpful for small start-ups wishing to penetrate challenging overseas markets or for larger firms intending to build innovative products at the lowest possible cost.

Alliances are forged at many different levels and for different reasons. The much publicized IBM-Apple alliances exemplify concerted attempts to create new industry standards from which both firms may potentially benefit. The relationship which Apple has forged with third-party suppliers is aimed at developing a wide range of products to fit several specialized market niches. Sun Microsystems benefited from early investments by Kodak and AT&T, which

(among other things) legitimized them in the eyes of wary customers unused to purchasing equipment from new firms.

The Firm: Barriers to Permanence

Swift action, mobility, and ephemerality are critical features of the Silicon Valley business dynamics. Organizational structures and business models undergo frequent strategic recalibrations and constant realignments. The ability of a firm to endure is thus contingent on the continued viability of its business model and its organizational capabilities. "Today's" successful models may only endure for a limited time. A firm must either recalibrate its approach or find a new business proposition, before it is made obsolete. In these conditions, *it is often easier to form a new entity in order to capitalize on a technological breakthrough or an unfolding market niche, since rapid concentration of effort is critical for success.*⁴⁸ A start-up firm has a singular focus and minimal inertia, whereas established companies—despite financial resources and market presence—can take longer to maneuver and recalibrate their business trajectories. There are several underlying forces which drive change and ephemerality in high-technology companies.

Organizational Alignment and Speed of Maneuver

The founders and the senior management team of a start-up are typically the product's developers, manufacturers, and marketers, fusing together thinking and doing, and aligning the organization behind its strategic objectives. As a firm grows, sometimes by as many as hundreds of people per month, its organizational structure can become quickly fragmented. Turf battles invariably arise which can fracture the cohesion of an organization, resulting in a distancing of its generals from the troops. Functional segmentation and compartmentalization, coupled with organizational inertia, can hinder the recalibration of business models, resources, and capabilities.⁴⁹ The resulting "loosely coupled" organization may be too cumbersome for rapid maneuvering. The point to note is that inertial forces restrict a firm's flexibility—its ability to change course and maneuver quickly as technological and market conditions change.⁵⁰

Employee Commitment and Motivation

Typically, employees in a start-up possess enthusiasm and motivation and enjoy the challenge of survival and personal growth. In contrast, employees of firms that have become an established public company value security and a relatively stable work environment. When a start-up becomes a public company or is acquired by another entity, many technical professionals—motivated by the opportunities for personal growth and the potential for significant financial gain—change their place of employment and forego the relative security of a large entity for the turmoil and sense of adventure associated with a new start-up.

Incentive and compensation structures and the “political” climate of established firms can also create a chasm between individual motivations and organizational goals. In a start-up, much of an individual’s substantive compensation is in the form of stock options—aligning individual and organizational goals, and generating intense commitment. Since technology industries are knowledge-intensive and knowledge resides in people, individual attitudes and motivation levels can make or break a firm and affect its position in an industry.

Maturing firms may have difficulty matching the cultural intensity of a start-up which is critical for building a team spirit and focusing emotional and creative energy on achieving the desired goal. Cultural intensity dissipates as a firm grows and becomes functionally segmented, potentially leading to a lost sense of purpose and confusion about the desired cultural values. These can have a disquieting impact on many professional employees and lead to a sense of apathy and loss of a community spirit.

Often, that initial cultural intensity fades when the founders leave and the original mission is accomplished.⁵¹ Large technology firms that have managed to retain their entrepreneurial atmosphere and creative spirit often have founders who stay with the firm for a long period of time—maintaining the connection with, and motivating the troops.⁵² In these situations, the founders can inspire the employees over time, evolving a shared set of strategic and cultural beliefs that bind the different parts of the organization and focus their collective energies on a common sense of purpose.⁵³

The Credibility of Newness

Another barrier to permanence in certain segments of high-technology industries—such as software—is the relative ease with which new firms can enter an arena. This process is facilitated since start-up firms in knowledge-based industries are less susceptible to problems associated with “liabilities of newness.”⁵⁴ Many start-ups can circumvent potential “legitimacy problems” by developing a revolutionary product or a service that fulfills a market need in a timely manner. This is often the case when a new product becomes a *de facto* standard because of its market impact and the speed with which it is adopted. Recent examples include Intuit’s “Quicken” software for home banking, Auspex’s network file server, and Verifone’s transaction automation systems.

Indeed, in some cases it may be argued that early legitimacy can be a liability because it may set unrealistically high expectations.⁵⁵ Silicon Valley has had its share of technology start-ups that had a great deal of initial legitimacy (in terms of the founding teams’ background, venture capital funding, favorable press reviews), but failed to deliver successful products and have subsequently gone out of business or were prematurely acquired.⁵⁶

There are many barriers to permanence in the fast-moving environment of Silicon Valley. First, the scientific know-how and technical expertise on which products are based are rendered obsolete by the constant creation of new knowl-

edge. Markets and distribution channels for technology products and services are fluid, as customers become familiar with the capabilities of products. Induced by the “renegade intent” (to transform and disturb the existing *status quo*) of start-ups and by the entry of large global corporations—competitive pressures invariably catch incumbents in a pincer movement. Just as with a football team or a rock-band where the “chemistry” between the players is critical, the entrepreneurial teams that found most firms are seldom enduring.

The following cases illustrate many of these forces in the context of two companies that we studied in some detail. These pioneering firms were well-known innovators in their respective industries and have ceased to exist in their original form. Their evolution demonstrates the *modus operandi* of Silicon Valley firms and the dynamic changes that drive the process of flexible re-cycling in the context of the ecosystem.

Case Histories

*Shugart Associates*⁵⁷

Shugart was incorporated in February 1973. It was founded by a team of 11 engineers, many of whom had previously left IBM to form Memorex. Funded with venture capital backing, the firm was initially engaged in the parallel development of a printer and a central processing unit, before narrowing its focus to a single business—floppy disk drives.

The company’s first product, an 8” floppy disk drive, was introduced in May 1973 with volume shipments beginning in July 1973. Abandoning the earlier “grand” strategy, the firm concentrated all of its resources on disk drives, resulting in the introduction of “double density” 8” floppy disk drives in April 1975. This was followed by another innovation which has subsequently become part of Silicon Valley folklore, the SA400 5¼” floppy disk drive which was introduced in September 1976.⁵⁸ This product was a major success propelling the company’s growth, with revenues rising to \$18.14 million by 1977. Shugart subsequently augmented its product line by developing an 8” Winchester disk drive, a process which, like that of the SA-400, was funded by two of its customers.

In 1978, Shugart postponed its initial public stock offering due to inauspicious trading conditions on Wall Street. It was subsequently acquired by Xerox Corporation—as part of its office automation strategy. Fueled by increasing demand for the 5¼” floppy and other product lines, Shugart’s revenues continued to grow rapidly. Following the acquisition, however, several members of its management team and engineering staff left the company—some to form “niche” disk drive start-ups, others to join competing firms. Over time, these spin-off firms were to play an almost-patricidal role in the demise of Shugart.

Until 1981, not yet weakened by these spin-offs and Japanese competition, Shugart continued to innovate and prosper. It pioneered, among others,

the Shugart Associates Standard Interface or SASI, so that systems integrators could quickly develop the microcode to allow Winchester disk drives to meet the unique operating systems requirements of different computer system manufacturers. Eventually, this product became an industry standard, enabling the interchangeability of disk drives between different and previously incompatible computer systems. SASI was the forerunner of the now-ubiquitous SCSI standard⁵⁹ which “controls” both IBM PC (along with PC-compatibles) and Apple Macintosh computers. During this period, Shugart was also engaged in other innovative product development activities, such as optical disk drive technology. In this case, a subsidiary unit, Optimem, was established as an international strategic alliance between Xerox and SGS Thomson of France.

Between 1981 and 1984, Shugart’s revenues grew considerably, although during this period, the industry and competitive dynamics were also transformed. Several niche spin-offs, together with other start-ups, began to compete with selected parts of its business. Furthermore, the Japanese manufacturers had made aggressive inroads into the low-end floppy disk drive segment—competing on the basis of low price and high quality. Yet Shugart Corporation (as it was subsequently renamed) was still, at this stage, the leading “across-the-board” supplier of rotating memory products.

Following an abrupt industry downturn which resulted in two years of consecutive losses, Shugart was disengaged by Xerox in 1985/86. The floppy disk business, which at the time constituted about one third of Shugart’s revenues, was sold off to its Japanese joint venture partner, Matsushita. Its Winchester disk drive division was dissolved and its optical disk drive unit, Optimem, was sold to Cipher Data, which itself was subsequently acquired by Archive.⁶⁰ The much-paired down Shugart, stripped of its 5¼” and pioneering 3.5” floppy and Winchester lines, was sold off to a group of investors, and continues to operate as a service and distribution center in Los Angeles, where today it concentrates on drive-refurbishing and end of life products.

ROLM Corporation

ROLM was founded in 1969 by four former student colleagues—Richeson, Oshman, Loewenstern, and Maxfield—who had known one another first at Rice University and later at the Stanford Engineering School. The founding team’s first business venture was to pioneer a unique commercial approach in selling minicomputers to the tradition-bound military establishment. Although the team established a viable business base in this niche market, their overriding objective was to build a sizable “commercial” concern. Minicomputers had been a timely opportunity to get the business off the ground. After three years, the founders became concerned about the limited size of that market and its potential for building a sizable commercial enterprise.

In the early 1970s, the founding team embarked on a preemptive search for a new business opportunity and decided to enter the telecommunications business. The 1968 “Carterfone” decision of the Federal Communications Com-

mission had partially deregulated the industry, thereby opening up the vast telephone equipment market to a host of new companies. The intention was to leverage ROLM's core technological skills by developing a computer-controlled telephone switching system, with enhanced capabilities compared with the traditional electromechanical units.

During the 1970s, ROLM had to be steered through the uncertain and stormy seas which characterized the industry. Computers were changing the nature of telephones and telecommunications. There was considerable debate and uncertainty about the eventual de-regulation of the industry, despite AT&T's concerted efforts to retain its long-standing position as a regulated monopoly. The interconnect distribution channel was just beginning to get off the ground, and competition had intensified with the entry of Japanese and European giants in the field. In short, the industry was in a state of chaos and confusion.

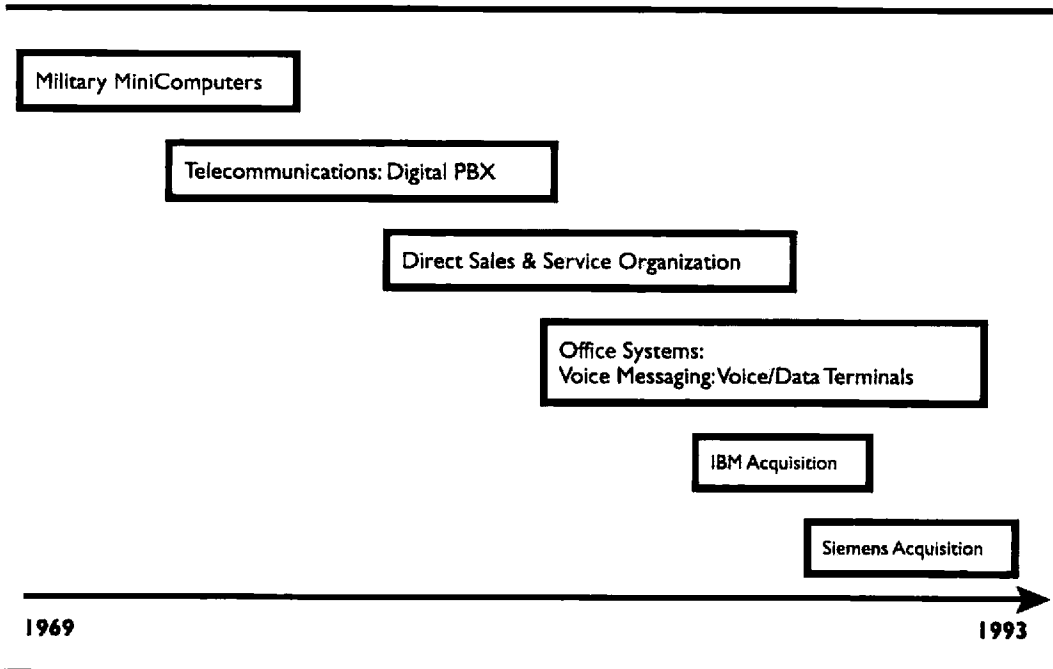
At the time, many expert observers questioned the ability of a young, unknown player, like ROLM, to survive, let alone prosper, in the face of such uncertainty. Despite the initial grave misgivings of many expert observers, ROLM managed to become a leading telecommunications firm during the next 5 years. By the late 1970s, telecommunications products accounted for almost 70% of its total revenues.

ROLM consolidated its strategic position during the late 1970s by setting up a direct sales and service organization, and it recalibrated its business trajectory during the early 1980s by making selected forays into the "office of the future"—focusing on integrated voice/data terminals and voice messaging systems. It was later acquired by IBM in the aftermath of AT&T's divestiture in 1984 and was partly sold to Siemens, the German giant, during the late 1980s.

At the time of the IBM acquisition, ROLM was called "the ship that is creating the wave of innovation in the field" and a "forerunner in the fast-paced (telecommunications) market."⁶¹ These tributes were clearly reflected in its impressive market performance. After only ten years in the telecommunications business, ROLM had managed to capture 15% of the market for office telephone switches, only 9% behind AT&T. Another measure of its remarkable success was the dramatic increase in its stock price. Compared with the 1,920 companies that have gone public since 1975, ROLM's stock has produced the largest long-term relative gain over its initial offering price.

Discussion and Observation

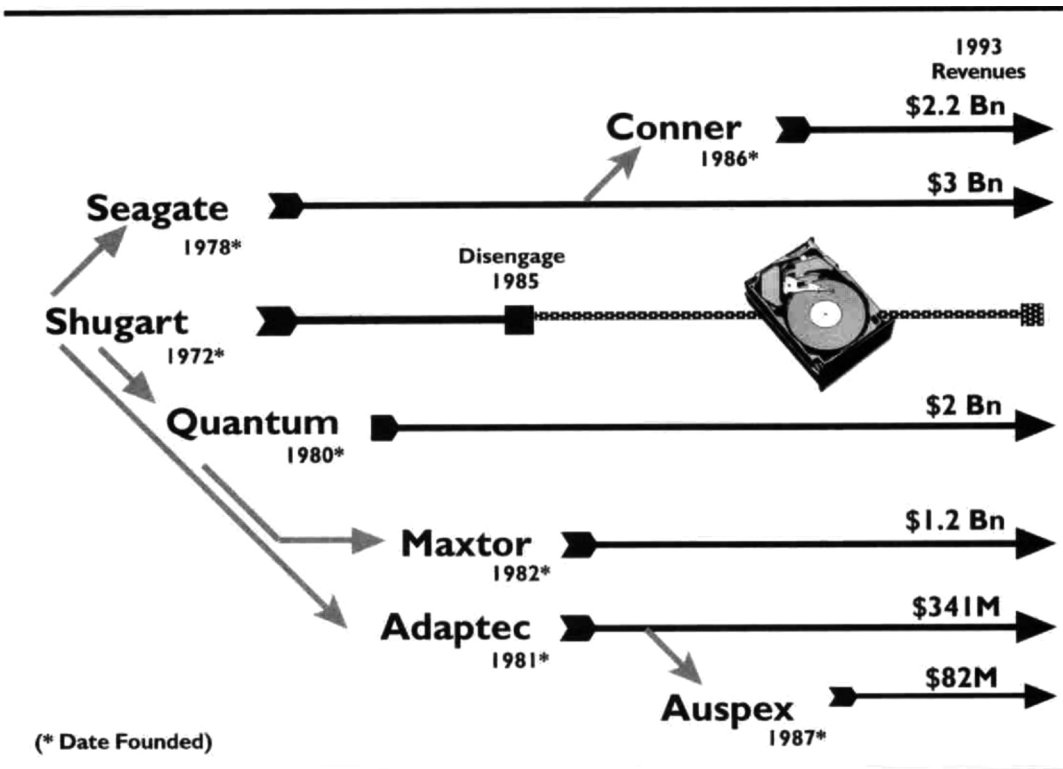
As these two cases illustrate, a high-technology firm's strategic trajectory is not sustainable in its original form. Shugart's initial mission, for example, was to develop a complete desktop computer system—including a printer, CPU, and disk drives. As indicated in Figure 3, ROLM's business model evolved through several stages during its fifteen years of life as an independent company. Even its original business plan included four different product ideas—none of which included the digital PBX that later fueled its growth.

FIGURE 3. Recalibrations of ROLM's Business Model

In order to survive, business models and product offerings must be perpetually re-calibrated.⁶² This ability to maneuver once a strategy is recalibrated is the hallmark of success and a necessary condition for survival.⁶³ Operating in conditions of kaleidoscopic change, a firm's technical demise, we suggest, is not necessarily detrimental as long as the ecosystem can flexibly re-cycle its critical know-how and human assets. The ephemeral nature of ROLM and Shugart clearly resulted in a re-configuration and re-cycling of their assets and capabilities. Many of Shugart's employees joined its spin-offs,⁶⁴ started new companies, and in some cases re-vitalized other entities.⁶⁵ Figure 4 sets out the genealogy of firms that spun out of Shugart during the late 1970s and 1980s. Similarly, many of ROLM's original employees, including their founding team, have since moved on to other entities, started a number of new concerns, and turned around a number of languishing companies.⁶⁶

Conclusion

The constellation of high-technology firms, venture capitalists, support infrastructure, globally diverse talent pool, and academic and research institutions which have coalesced into Silicon Valley characterize a dynamic ecosystem of semi-autonomous yet mutually supportive entities, communities, and cultures.⁶⁷ It has generated a critical mass of global, knowledge-intensive enterprises, whose innovative products and technologies are transforming the global

FIGURE 4. Spin-offs in the Disk Drive Industry

economy. Once the ecosystem, in which these firms operate, reached a critical mass, the evolution of existing firms, and the creation of new entities, produced a dynamic domain in which the process of firm formation, demise, and re-cycling occur on a continuous basis.

Paradoxically, this ecosystem enables continuity and change to co-exist. It thrives on processes of flexible recycling, as start-ups and spin-offs are formed, and on continuous recalibration, as incumbent firms re-assess their business models. Flexibility, "the ability to do something differently or do something other than that originally intended,"⁶⁸ is a *sine qua non* for high-technology firms, since conditions are rarely stable enough to enable, what economists term, "perfect adaptation."⁶⁹

This process of "flexible recycling" is enhanced in the absence of the typical stigma associated with organizational failure. Indeed, the high failure rate may be more of an asset in that it increases the rates of experimentation and speed of recycling, and in a Popperian sense,⁷⁰ entrepreneurs learn, just as scientists do, much from failed experiments. Since organizational death, in and of itself, is not perceived as a finite expression of failure, entrepreneurs are able to entertain what would normally be considered "outlandish" risks. Moreover, prospects of failure and ephemerality can also reduce feelings of over-confidence

and invulnerability among successful incumbents and keep them on their toes. In such a setting, large incumbent firms strive to become “Agile Giants”⁷¹—capable of rapid maneuvering and recalibrating.⁷²

The short life cycle of many high-technology firms may be helpful for sustaining the long-term innovative capability of an ecosystem such as Silicon Valley. In addition to maintaining the stream of new firms, which in turn provide employment opportunities and create new products and services, ephemeral firms increase the variety of experiments—and, when acquired, they can help rejuvenate other entities or become re-configured in the form of new entities.⁷³ This variety can also facilitate the development of the next generation of firms and products. This is crucial, given that it is difficult, if not impossible, to know the “winning formula” *a priori*.

The notion of an ecosystem that thrives on flexible recycling, may provide an alternative vehicle for policy-makers to examine issues related to global competitiveness and the growth of high-technology industries. National and regional governments and local communities may become as concerned with the long-term success of their ecosystems as they are with the survival of larger firms which provide substantial employment opportunities in their communities. In thinking about these considerations, however, the appropriate balance needs to be struck between anchors of employment stability and growth and the flexibility needed to accommodate kaleidoscopic changes.

It is clear that the managerial recipes and organizational regimes of many Silicon Valley companies are worthy of further study and discussion. In addition to the nimble start-ups, the mid-sized “adolescents” and large enduring firms must also be capable of continuously recalibrating their trajectories and reconstituting their business models. They must become “agile giants” to survive in this dynamic ecosystem. However, such firms demonstrate that it is indeed possible for large, mid-sized, and small firms to be flexible as long as their organizational systems are *aligned* and their strategies are *recalibrated* as conditions they confront inevitably change, thereby rendering them capable of rapid *maneuver*.

Permanence or sustainability may not be the most critical measure of success in this fast-moving and knowledge-intensive domain.⁷⁴ Other considerations include the extent to which a firm can pioneer new products and services, generate wealth for its shareholders, and foster enduring relationships among its employees. However, most of our managerial tools, conceptual frameworks, and research studies—reflected in managerial expectations and business theories—are framed in the context of sustainability and the rhetoric of permanence. As Carol Bartz, the CEO of Autodesk, observed:

We need to constantly re-evaluate our business models and our industry forces, and take corrective measures along the way. Yes, it's part of natural business evolution, but it also signals the end of an era.⁷⁵

ACKNOWLEDGEMENTS

Over the past decade, we have benefitted from discussions with many entrepreneurs and professionals working in Silicon Valley, in addition to several scholars. Although it is impractical to mention everyone by name, we wish to extend our sincere thanks to the following individuals for their insights and contribution to our research: Larry Boucher, Glenn Carroll, John Child, Deborah Engel, Jim Guzy, Trip Hawkins, Gunnar Hedlund, Roger Lang, Guiliana Laven-del, Hal Leavitt, David Lee, Gianni Lorenzoni, Giacomo Marini, Bob Maxfield, Bob Metcalfe, Dick Moley, Merrill Newman, Ken Oshman, Dennis Paboojian, Jim Patterson, Ted Pittiglio, John Rollwagen, George Sollman, Kevin Sullivan, David Tsang and Hatim Tyabji . We are solely responsible for any errors.

References

1. This is reflected in the comments of a high-tech executive on changes in the disk drive industry: "Less than a year ago the 20 Megabytes (MB), 2.5" disk drive was the standard product. It has since moved to 40, and 60, and now 80 MB. The most desirable product for the marketplace has basically gone through 3 to 4 product generations in the last 12 months" Roger Leisy, Director of Sales, PrairieTek Corp., Electronic Buyers' News, September 21, 1991, p. 14.
2. By 1993, some 18 firms had achieved revenues in excess of \$1 billion, and overall approximately ¼ million jobs and 2,150 high-technology companies have been created in Silicon Valley. *San Jose Business Journal*, November 30, 1992, p. 6.
3. According to the Center for the Continuing Study of California, California Employment Development Department (1991), Silicon Valley's employment profile comprised 69,900 jobs in business services, 67,400 in electrical components, 57,500 jobs in the computer sector, 31,300 in electronic instruments, 31,300 in information and software, 18,100 in communications, and 5,800 in biotechnology.
4. H. Bahrani, "The Emerging Flexible Organization: Perspectives from Silicon Valley," *California Management Review*, 34/4 (Summer 1992).
5. In a recent article, Peter Drucker suggests that many of the assumptions on which organizations have historically been built and managed, no longer fit the emerging reality. See P. F. Drucker, "The Theory of the Business," *Harvard Business Review* (September/October 1994).
6. See, for example, J. Williams "How Sustainable is Your Competitive Advantage?" *California Management Review*, 34/3 (Spring 1992).
7. See M. Hannan and J. Freeman "The Population Ecology of Organizations," *American Journal of Sociology*, 62 (1977): 929-964.
8. There are parallels with Schumpeter's notion of "creative destruction." See J.A. Schumpeter, "An Analysis of Economic Change," *The Review of Economic Statistics*, Volume 17 (May 1935).
9. Well-known firms which have been discontinued include Visicorp, the spreadsheet software pioneer; Osbourne Computer, which pioneered the portable computer; Xebec, the developer of I/O controllers; and Momenta, the first pen-based computer company.
10. G. Owen and L. Kehoe, London's *Financial Times*, June 18, 1992 p. 16.
11. Their "ephemeral" nature does not imply the dissolution of all their know-how and intellectual capital. In the disk drive controller industry, for example, the

- demise of Xebec had a positive impact on the growth of Cirrus Logic, as did the disengagement of Scientific Microsystems, which led to the resurrection of SuperMac Technologies as an independent firm.
12. It is interesting to note that, even after a brief period, several of the "excellent" companies described in T. Peters and R. Waterman, *In Search of Excellence* (New York, NY: Harper & Row, 1982), when re-examined were found to have slid from their pre-eminent positions. See "Who's Excellent Now," *Business Week*, November 5, 1984, p. 76:.
 13. According to a recent study, smaller organizations are becoming more dominant in the American economy. "[Their] numbers have increased disproportionately in recent years...the typical or average American organization is much smaller than it was in the 1960s." G. R. Carroll, "Organizations: The Smaller They Get," *California Management Review*, 37/1 (Fall 1994).
 14. Well-known industry-focused studies include J. Freeman, "Ecological Analysis of Semiconductor Firm Mortality," in J.V. Singh, ed., *Organizational Evolution: New Directions* (Newbury Park, CA: Sage Publications, 1990); and G. R. Carroll and J. Delacroix, "Organizational Mortality in the Newspaper Industries of Argentina and Ireland," *Administrative Science Quarterly*, 27 (1982): 169-198.
 15. In recent years, there has been a growing interest in Silicon Valley in the context of regional development and the phenomenon of the "technopolis." See E.M. Rogers, Ying-Chung, and Annie Chen, "Technology Transfer and the Technopolis," in M.A. Von Glinow and S.A. Mohrman, eds., *Managing Complexity in High-Technology Organizations* (New York, NY: Oxford University Press, 1990); and AnnaLee Saxenian "Regional Networks and the Resurgence of Silicon Valley," *California Management Review*, 33/1 (Fall 1990).
 16. Drawing on these field studies, the organizational and strategic features of high-technology firms have been elaborated elsewhere. See H. Bahrami and S. Evans, "Stratocracy in High-Technology Firms," *California Management Review*, 30/1 (Fall 1987); H. Bahrami and S. Evans, "Strategy-Making in High-Technology Firms: The Empiricist Mode," *California Management Review*, 31/2 (Winter 1989); H. Bahrami and S. Evans, "Emerging Organizational Regimes in High-Technology Firms: The Bi-Modal Form," *Human Resource Management Journal*, 28/1 (Spring 1989); S. Evans "Strategic Flexibility for High-Technology Maneuvers: A Conceptual Framework," *Journal of Management Studies*, 28/1 (January 1991).
 17. For a detailed discussion of the evolution of industries in different national contexts see M. Porter, *The Competitive Advantage of Nations* (New York, NY: Free Press, 1990).
 18. See "Hot Spots," *Business Week*, October 19, 1992, p. 80.
 19. Hsinchu was initiated in 1980 and has attracted an investment of \$2.5 billion to date. A total of 145 firms, over half of which are Taiwanese, generated revenues of around \$3.3 billion in 1992. Of these, 40 firms are in semiconductors, 46 in computers and peripherals, 24 in telecommunications, and 15 in electro-optics. *San Francisco Examiner*, October 17, 1993.
 20. See E. Garnsey and A. Cannon Brookes, "The Cambridge Phenomenon Re-Visited," Research Paper, Judge Institute of Management Studies, Cambridge University, September 1992.
 21. Addressing a meeting of the Churchill Club (October 1992), Bill Hewlett suggested that the origins of Silicon Valley, in his opinion, almost date back to the development of ship-to-shore radio and the early days of television—before RCA moved its R&D labs to the East Coast.
 22. One of the driving forces underscoring these extraordinarily high growth rates is the structure of Venture Capitalist's limited partnerships which typically last for 7 years. In order to show a return to the shareholders, investors need to liquidate

their positions within the time frame specified by the limited partnership. Therefore, the exit strategy of a portfolio company, an initial public offering or an acquisition by another firm, needs to be orchestrated within this time frame, so that the proceeds can be shared among the partners and the investors. A general rule of thumb applied by Venture Capitalists is to evaluate a deal on the basis of achieving 5 times the value of their initial investment in 3 years or 10 times in 5 years. These rates of return may appear to be abnormally high, but when, as a rule of thumb, 1 in 20 firms actually getting off the ground and even fewer going public, they can provide part of the motivation behind rapid growth. Venture Capitalists often hedge their bets by investing in several firms vying for dominance in any given arena. Even the most seasoned venture capitalists may be unable to spot winners early on—as witnessed by their initially unfavorable reactions to ROLM, Seagate, and Adaptec, among many other successful ventures, or indeed their favorable disposition towards Plexus, Momenta, and EO.

23. J.M. Harker, D.W. Brede, R.E. Pattison, G. R. Santana, and L.G. Taft, "A Quarter Century of Disk File Innovation," *IBM Journal of Res. Development*, 25/5 (September 1981).
24. The team comprised, among others, L.D. Stevens and Ken Houghton (later the Dean of Engineering at Santa Clara University).
25. L.D. Stevens "The Evolution of Magnetic Storage" *IBM Journal of Res. Development*, 25/5 (September 1981): 663-675. For an evolutionary perspective on the production of such devices, see R.B. Mulvany and L.H. Thompson, "Innovations in Disk File Manufacturing," and K.E. Haughton, "An Overview of Disk Storage Systems," both in *Proc. IEEE* 63, 1975.
26. The U.S. disk drive industry is dominant in merchant production, although much of its manufacturing is conducted offshore, particularly in Singapore, Malaysia, and Thailand, or in conjunction with Japanese partners. The industry remains vibrant, as attested by the sustained growth rates and the pace of innovative new product development.
27. The first tenant was Varian Associates—a spin-off from the University Physics department. Hewlett-Packard followed in 1954. David Packard and William Hewlett, Terman's former students, had founded Hewlett-Packard in 1939 to commercialize an electronic device called a "variable frequency oscillator" which Hewlett had developed as part of his graduate work.
28. See E. Rogers and J. Larsen *Silicon Valley Fever*, (New York, NY: Basic Books, 1984); and M. Johnston "High Tech, High Risk, and High Life in Silicon Valley," *National Geographic*, 162/4 (October 1982): 459-476.
29. See J. Kotkin and P. Grabowicz, *California, Inc.* (New York, NY: Rawson, Wade Publishers, 1982).
30. For example, Bill Miller, former president of SRI, was also a professor at the Stanford Graduate School of Business.
31. When the disk drive company, Seagate, was founded by Alan Shugart in 1979, venture capitalists were not initially very interested; private investors—including Norman Dion, the founder of Dyson, and later TI and Compaq's Rod Canion provided the seed money to get the firm off the ground. Later, when Finis Conner wanted to start a new disk drive firm, at a time when that business was out of favor at Sand Hill Circle (where many venture capitalists were based), Canion again stepped in and made an investment whose pay-off was substantial. Larry Boucher and his co-founders took their initial business plan for starting Adaptec, the input/output controller company, to venture capitalists in vain. They mortgaged their houses in order to build a prototype of the product before venture capitalists invested in the company. Similarly, ROLM's founding team initially sought venture capital backing without much success. Each of the four founders

- invested \$15,000 in the business, before Jack Melchor—one of Silicon valley's early venture capitalists—agreed to become a fifth founder by investing \$15,000, and guaranteeing a bank line of credit for \$100,000 in order to get the firm off the ground.
32. Departing from the traditional mode of venture funding, a recent trend has been the investment by larger firms in high-technology start-ups. EO Corporation, General Magic, and 3DO were initially funded in this manner.
 33. Venture capitalists seldom invest alone in a deal, but with a group of other venture capital firms. For a historical description of the role of venture capital in Silicon Valley see W. R. Hambrecht "Venture Capital and the Growth of Silicon Valley," *California Management Review*, 26/2 (Winter 1984).
 34. In recent years, many firms including 3Com, Sun Microsystems, Silicon Graphics, Conner Peripherals, and Syntex have built campus-like settings in order to generate a community spirit and to facilitate inter-group communication and interaction.
 35. Well-known examples include Italians (Frederico Fagin, co-founder of Synaptics and Zilog, Giacomo Marini co-founder of Logitech), French (Jean Louis Gasse, formerly of Apple, and Phillippe Kahn of Borland), British (Wilf Corrigan of LSI Logic, Dick Moley of Stratacom), Chinese (David Lee of Qume), Hungarian (Andy Grove and Les Vadacz of Intel), Indian (Juggy Tandon of Tandon), German (Michael Spindler of Apple Computer), Iranian (Kamran Elahian of Cirrus Logic and Momenta), Israeli (Arieh Feingold, the co-founder of Daisy Systems), and other nationalities.
 36. J.A. Schumpeter "The Creative Response in Economic History," *Journal of Economic History*, 7 (November 1947): 149-159.
 37. See *San Jose Mercury News*, Business Monday Venture Tree Quarterly Survey, November 29, 1993.
 38. See C.C. Swanger "Apple Computer: The First Ten Years," Case Study #S-BP-245, Graduate School of Business, Stanford University, 1985.
 39. Interestingly, Don Massaro, who replaced Alan Shugart as President of Shugart, became VP of Marketing at Conner Peripherals, after co-founding Metaphor Computers with Dave Liddell, co-inventor of Ethernet at Xerox PARC.
 40. See L. Kretchmar "Auspex Serves Notice," *Upside Magazine* (November/December 1989), p. 17-18.
 41. *San Jose Business Journal*, July 13, 1992, p.14.
 42. A good case in point is Gordon Campbell, the founder of SEEQ Technology, who after leaving SEEQ went on to found Chips and Technologies. C&T managed to become one of the fastest growing technology companies during the late 1980s. Similarly, Al Shugart—after leaving Shugart Associates—went on to found the successful Seagate Technology, the pioneer of 5.25" Winchester disk drives.
 43. As pointed out by other researchers, the phenomenon of the "Living Dead" illustrates the futility of maintaining a high-technology firm, which has not illustrated demonstrable success, on an expensive life support system. See L.J. Bourgeois and K.M. Eisenhardt, "The Anatomy of the Living Dead in the Computer Industry," *California Management Review*, 30/1 (Fall 1987).
 44. M. A. Maidique and B.J. Zirger, "The New Product Learning Cycle," *Research Policy* (December 1985).
 45. See Bahrami and Evans (Winter 1989), op. cit.
 46. The notion of "diverse specialization" was discussed in M. Piore and C. Sabel, *The Second Industrial Divide* (New York, NY: Basic Books, 1984).
 47. In this regard, global and overseas companies have shown a great deal of interest in investing and forging alliances in Silicon Valley companies. See A. Beshar "Asian Investor Feast," *Upside Magazine* (November/December 1989); "Foreign

Investment in Bay Area Bioscience," a survey prepared by Booz-Allen Hamilton for the Bay Area Bioscience Center, 1991; D.J. Teece "Direct Foreign Investment in Silicon Valley," *California Management Review*, 35/2 (Winter 1992).

48. See, for example, General Magic and 3DO as recent examples.
49. Scott McNeally described how Sun Microsystems's organization overcomes this problem by being both "aligned yet loosely couple." See "Interview with Scott McNeally," *Upside Magazine* (March 1993).
50. This phenomenon is clearly observed in the disk drive industry. Over the last decade or so, the firms which have dominated the market for a particular "form factor" (the technical term for the size of the drive—which has evolved in ten years from 8 to 5.25 to 3.5 to 2.5 and recently to 1.8 inches) typically have been late to market with the next generation device. For example, Seagate, the pioneer of 5.25 inch Winchester disk drives (which grew at a tremendous pace during the early 1980s with the rapid take-off of the personal computer market), initially failed to capitalize on the emerging opportunities presented by 3.5 inch disk drives destined for the portable computer market. However, Conner Peripherals—a new start-up founded in 1986 by Finis Conner, a Seagate co-founder and former senior executive—became a major success story in the late 1980s by developing the 3.5 inch disk drive. Earlier, Quantum Corporation had a difficult time making the transition from the 8 inch line, of which it was a pioneer, to the 5.25 inch Winchester drives, which became a standard in the PC era. The firm learned valuable lessons from this experience and quickly became a substantial force in the emerging 3.5 inch form factor in the late 1980s. See J.L. Bower and C.M. Christensen, "Disruptive Technologies: Catching the Wave," *Harvard Business Review*, (January/February 1995).
51. For a discussion of the importance of corporate culture, especially in high-technology companies, see C.O'Reilly, "Corporations, Culture, and Commitment," *California Management Review*, 31/4 (Summer 1989).
52. Well-known examples include Intel, Hewlett Packard, Tandem, AMD, Sun Microsystems, Seagate, Conner Peripherals, Microsoft, and Oracle.
53. In certain cases, the founders may actually stay beyond their useful tenure, with an adverse impact on the firm's ability to recalibrate its business model as new realities unfold.
54. Recent studies have argued that new organizations have a high propensity to fail, partly due to problems of initial legitimacy—encapsulated in the term "liability of newness." See J.V. Singh, D. Tucker, and R.J. House, "Organizational Legitimacy and the Liability of Newness," *Administrative Science Quarterly*, 31 (1986): 171-193; H.E. Aldrich and C.M. Fiol, "Fools Rush In? The Institutional Context of Industry Creation," *The Academy of Management Review*, 19/4 (October 1994): 645-670.
55. A good case in point is Plexus Corporation, a high profile start-up which received venture capital funding of \$35 million. Another case in point is Stardent, formed through the merger of Ardent and Stellar.
56. A clear example is Trilogy, founded in the early 1980s by Gene Amdahl, who had earlier started Amdahl Corporation in 1970 in order to compete with IBM in the mainframe computer business. Trilogy attracted a lot of interest and venture capital in its formative years, but failed to survive because it could not deliver what was initially promised—an advanced microprocessor. Interview with Carlton Amdahl, *Upside* (February/March 1992).
57. For a more extensive review of Shugart and its industry context, see Case Study #S-MM-8, "Shugart Corporation: Planning Manufacturing Capabilities," and its accompanying Industry Note #S-MM-6N, "Disk Drives for Small and Microcomputer Systems, 1983," Graduate School of Business, Stanford University 1985.
58. For a synopsis of this episode, see Evans (1991), *op. cit.*

59. SCSI was pioneered by Adaptec, which was founded by three Shugart employees. When an internal business plan was submitted to Shugart's executive team, as part of a new internal venture funding initiative, it was turned down. As a result, the principal developers of this interface, Larry Boucher and two of his colleagues, left Shugart in 1981 to found Adaptec, an input-output controller company. By 1992, Adaptec had amassed revenues of \$311 million and was ranked among the top 200 public high-technology companies.
60. In 1992, Archive was acquired by Conner Peripherals, whose founder, Finis Conner, was a co-founder of Seagate and Shugart.
61. *San Francisco Chronicle*, September 26, 1984. For additional perspectives on the IBM/ROLM merger, see the *Economist*, September 29, 1984 and the *Wall Street Journal*, September 27, 1984.
62. The "Valley" is resplendent with examples of such recalibrations. Recent examples include Apple Computer, Hewlett-Packard, Oracle, Sun Microsystems, and Quantum Corporation. For further details, see "Now Comes the Real Battle at Oracle," *Upside* (May 1991), p. 32; "Back to Basics at Hewlett-Packard," *Upside* (June 1991), p. 38.
63. In many situations, the original intent seldom turns out to be the winning business proposition. For example, Shugart began by developing a complete desk-top computer including a printer, before it concentrated on disk drives; ROLM's first business plan included four different business ideas, none of which included the digital PBX, which later fueled its rapid growth.
64. Previous research has adopted a narrow concept of a spin-off, in part due to the traditional research focus on large corporations and their attempts at improving innovation. A spin-off is, under such circumstances, defined as one which is founded entirely by employees of the parent. In Silicon Valley most start-ups involve the participation of individuals from several different organizations. Unlike conventional analyses of spin-offs where entrepreneurs depart from a single parent, in Silicon Valley many new start-ups are founded by teams of entrepreneurs, oftentimes from different firms. Many of the companies which ROLM and Shugart helped spawn were not all necessarily direct spin-offs, in the limited sense of the term, in that employees joined forces with specialists in other areas to form new companies pioneering the next generation of technologies.
65. For example, in 1985 George Sollman (Shugart's former Vice President of Sales and Marketing) moved to Centigram—at the time, a struggling voice messaging company—where storage technology is a principal component of their system. In 1992, Centigram orchestrated a successful initial public offering, largely due to Mr. Sollman's ability to reincarnate their business model.
66. Ken Oshman, its founding Chief Executive, has moved on to lead Echelon, an emerging pioneer in the networking field. A number of former employees of ROLM's Mil-Spec Division have founded Ultra, a firm working on high-speed local area networks. Richard Moley, ROLM's former Vice President of Marketing and its first PBX product manager, is CEO of Stratacom, a manufacturer of multiplexers and network management systems. For additional details, see *San Jose Mercury News*, March 23, 1987.
67. The disk drive sector is somewhat self-contained, yet it depends on: several other specialist entities for its continued ability to innovate; integrated circuits for its electronic and I/O control units; thin film deposition technology for building the read-write recording heads; and the platters of media which can store data at ever increasing densities.
68. A.G. Hart, "Anticipations, Business Planning, and the Cycle," *Quarterly Journal of Economics* (February 1937), pp. 272-293.

69. As Stigler suggests, "adaptability" implies a singular and permanent adjustment to a newly transformed environment, whereas "flexibility" refers to successive, though temporary, adjustments that are made as conditions evolve. See G.J. Stigler, "Production and Distribution in the Short Run," *Journal of Political Economy*, 47 (1939): 305-327. In some cases, this recalibration may involve a migration of existing businesses, such as Shugart's migration from 8 to 5.25 and later to 3.5 inch disk drives; in others, as was the case at ROLM, it may entail the development of new businesses through the cross pollination of several technologies, such as digital computers and telephony.
70. K.R. Popper, *Conjectures and Refutations: The Growth of Scientific Knowledge* (London: Routledge & Kegan Paul, 1972).
71. As Dr. Andy Grove, CEO of Intel, was quoted in London's *Financial Times*, the agility comes in part from "newcomers sniping at their heels." G. Owen and L. Kehoe, *Financial Times*, June 18, 1992, p. 16.
72. There are several examples of large Silicon Valley firms which have successfully recalibrated their business models and have moved on to greater heights, including Intel, Oracle, and Hewlett Packard—just a few of the well known cases in recent years.
73. Several firms may be brought together in order to gain a critical mass. Macro-media, for example, was recently forged from three software companies all active in the burgeoning multimedia arena. MacroMind, Paracomp, and Authorware were merged together to set industry standards and orchestrate a successful initial public offering. Others examples include, Scientific Microsystems acquisition of OMTI (One More Time Inc.), Borland's acquisition of Ashton Tate, National Semiconductor's absorption of Fairchild, and Silicon Graphics' acquisition of MIPS.
74. Schumpeter argued that "the earning capacity of any industrial concern peters out after a time," J.A. Schumpeter, "Economic Theory and Entrepreneurial History," (1949), in R. Clemency, ed., *Essays on Entrepreneurs, Innovations, Business Cycles, and the Evolution of Capitalism* (New Brunswick, NJ: Transaction Publishers, 1991), p. 258.
75. Interview with Carol Bartz, *Upside* (April 1993), p. 44.